

Streaming Data Management and Time Series Analysis project

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1 Introduction

This project focuses on forecasting daily values for a period spanning from April 1st, 2015 to November 7th, 2015. We'll employ three different approaches: ARIMA modelling, UCM (Unobserved Components Model), and Machine Learning techniques. To evaluate our predictions, we'll primarily use Mean Absolute Error (MAE), while also considering Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) metrics specifically for the ARIMA and UCM models. Our historical data covers over eight years, running from January 4th, 2007 through March 31st, 2015. This project was done using Python language.

This dataset tracks how long it takes to resolve service requests in the US. The data is presented as a floating-point time series showing daily averages, with each data point representing how many days it typically took to complete requests that were closed on that specific day. The requests are similar to what you might see in technical support tickets. All values are positive numbers.

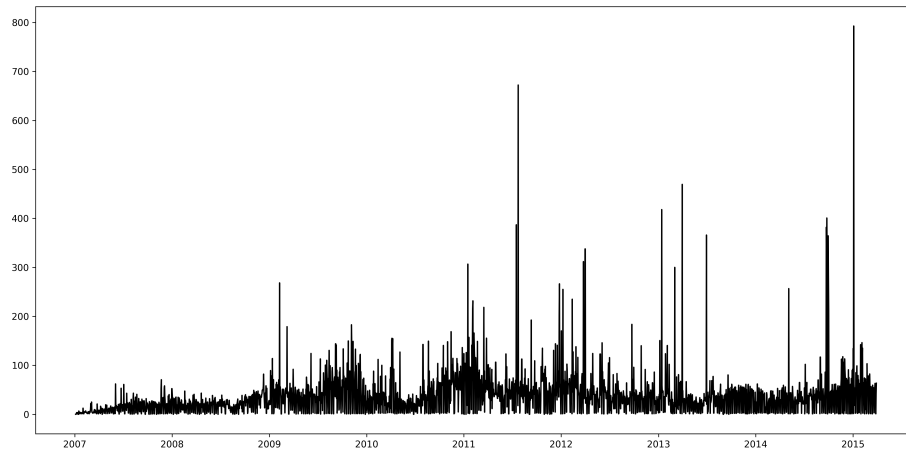


Figure 1: Distribution of the data

2 Preprocessing

For the preprocessing part, an analysis on missing data in our time series was conducted and the results can be seen in 2:

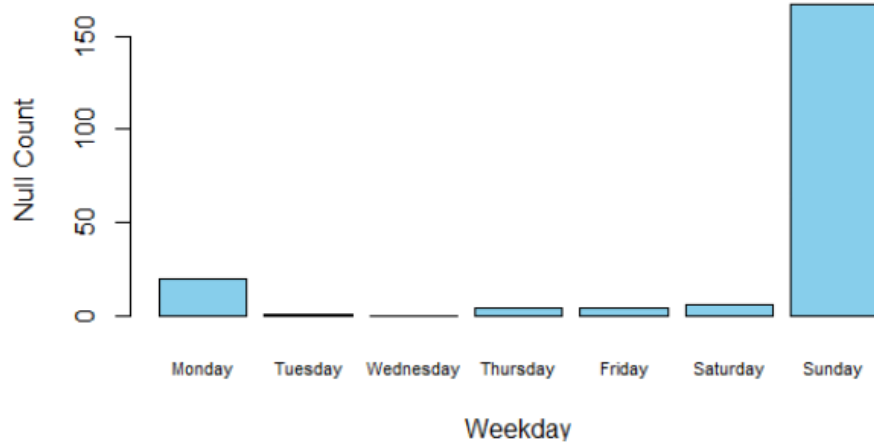


Figure 2: Null counts for each day

As we can see, there are exactly 202 NaN values, especially on Sunday. The technique used to solve this problem was a forward fill technique, that is replacing each NaN value with the last valid value before it. This is a simple and common imputation technique, assuming the last known value is a reasonable estimate for missing points.

The same issue was present with zero values in the time series, having 25 zero values. In order to manage this issue, we replaced these zero values with 0.00001, a value higher than zero but still so little that it wouldn't make any interferences with the distribution of our data.

The data presents a pattern characterized by numerous peaks and a core distribution where 50% of values lie between 9.5 and 48.0 days, as we can see in the boxplot 3.

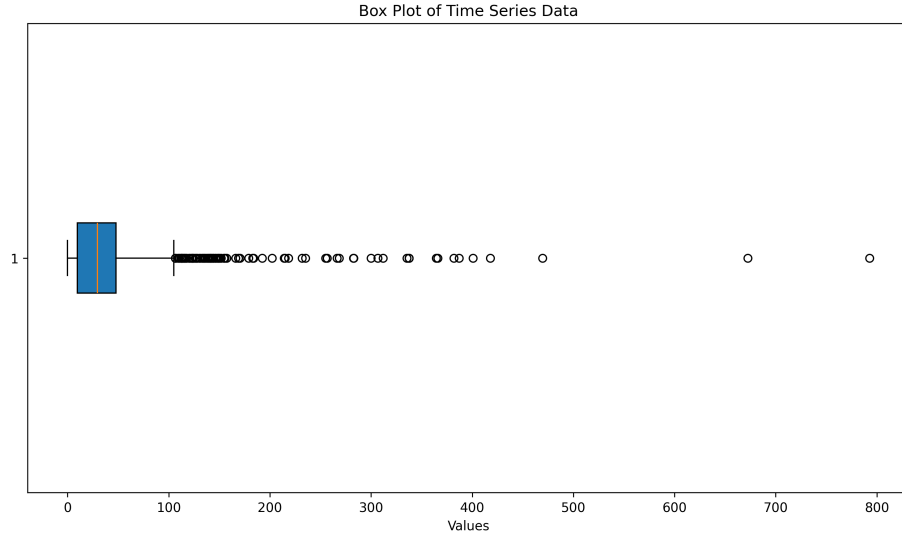


Figure 3: Boxplot of the data

To address the outlier issue evident in the boxplot 3, we implemented a robust statistical cleaning approach based on the interquartile range (IQR) method. The IQR, defined as the difference between the third quartile (Q3) and first quartile (Q1), is a measure of statistical dispersion. We identified outliers as data points falling outside the range $[Q1 - 1.5IQR, Q3 + 1.5IQR]$. Any observations beyond these boundaries, which were identified as -48.1056 and 105.6286, were removed from the time series to ensure more reliable analysis and forecasting.

After this outlier treatment, the resulting time series demonstrates clearer underlying patterns. The cleaned data reveals distinct seasonal fluctuations. This seasonality manifests as recurring patterns at regular intervals, suggesting systematic temporal variation in the data.

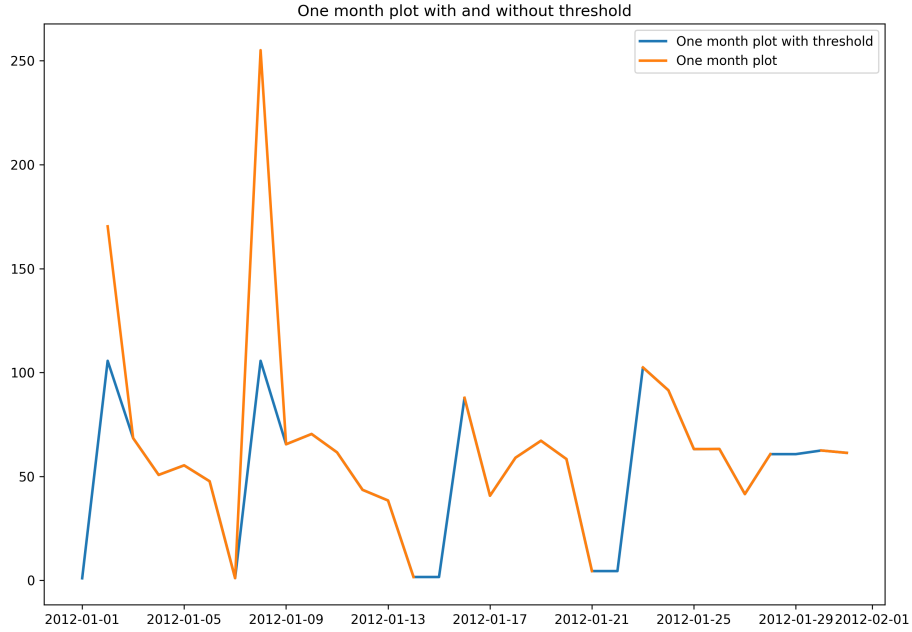


Figure 4: One month plot with and without outliers

3 Stationarity

After completing the pre-processing steps, we performed the Augmented Dickey-Fuller (ADF) test to examine the stationarity of the time series. Stationarity is a fundamental assumption for many time series models, indicating that statistical properties such as mean and variance remain constant over time. The ADF test evaluates the null hypothesis that a unit root is present in the time series, implying non-stationarity. In our analysis, the ADF test produced a statistic of -3.77 with a corresponding p-value of 0.0032, which is below the common significance level of 0.05. Additionally, the test statistic was lower than the critical values at the 1% (-3.43), 5% (-2.86), and 10% (-2.57) significance levels. Therefore, we rejected the null hypothesis and concluded that the time series is stationary, making it suitable for ARIMA and other time series forecasting models.

But after analyzing how the mean and standard deviation behave together we can see that these two values change over time, indicating that there is actually no stationarity:

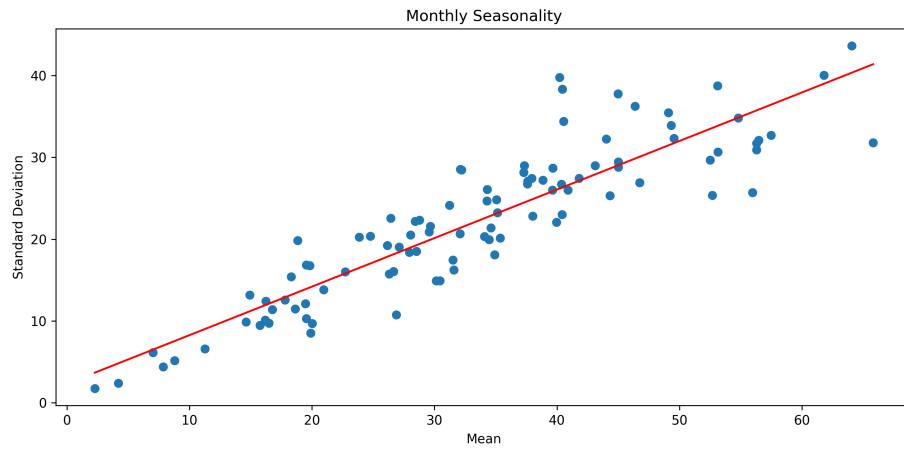


Figure 5: Monthly seasonality

One solution we found was to apply a BoxCox transformation to our time series, using a 0.15 lambda. This seemed to fix this problem, ensuring now a real seasonality:

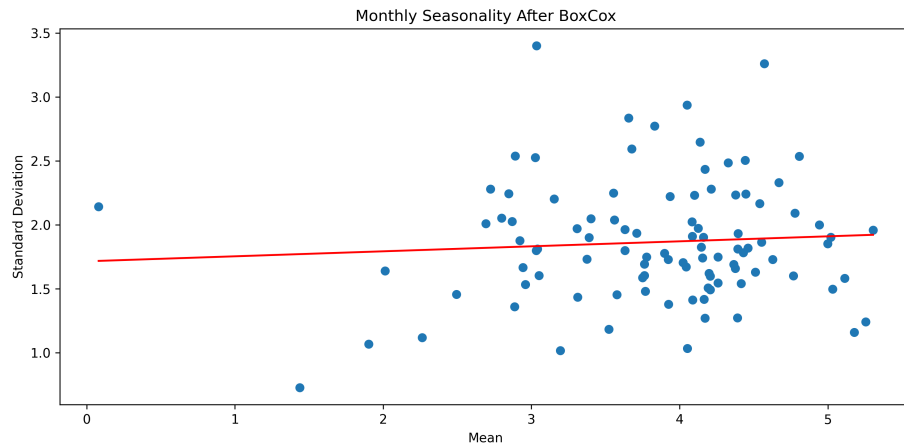


Figure 6: Monthly seasonality after BoxCox

4 ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a powerful statistical method used for time series forecasting. It models a time series based on its own past values (autoregression), the difference between past observations (in-

tegration), and past forecast errors (moving average). The goal of ARIMA is to capture the underlying patterns and predict future values while handling trends and seasonality effectively.

To identify the best ARIMA model, model selection criteria such as Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were chosen because they balance model complexity and goodness of fit. Mean Absolute Error (MAE), though, was the principal criterion for the best ARIMA model selection.

In this project, three ARIMA-based models were developed to forecast future values. The first model, $\text{ARIMA}(1, 0, 1)(2, 0, 1)[7]$, was chosen to capture both short-term autocorrelation and seasonal patterns. The non-seasonal components include one autoregressive (AR) term and one moving average (MA) term, selected based on significant lag structures observed in the partial autocorrelation function (PACF) and autocorrelation function (ACF) plots. The seasonal order $(2, 0, 1)$ with a period of 7 reflects weekly seasonality, with two seasonal AR terms and one seasonal MA term.

The second model, $\text{SARIMAX}(1, 0, 0)(1, 1, 1)[7]$, incorporates a first-order seasonal differencing step ($D=1$) to address residual non-stationarity after initial transformations. The simplified non-seasonal $\text{AR}(1)$ term was effective in capturing dependencies, as indicated by residual analysis and model fit metrics. The SARIMAX (Seasonal AutoRegressive Integrated Moving Average with exogenous regressors) model extends the traditional ARIMA model by providing enhanced flexibility for time series forecasting. Unlike ARIMA, which does not explicitly handle seasonality, SARIMAX includes additional seasonal terms (P, D, Q) with a specified period (s) to model repeating patterns. In this project, a period of 7 was chosen to capture weekly seasonality effectively.

Finally, the third model includes exogenous variables (weekend and holiday dummies) to enhance predictive power by accounting for external influences. The optimal orders for the exogenous model were determined using the *autoarima* function, which selected parameters based on AIC and MAE minimization. Model orders were validated by comparing AIC, BIC, and MAE across alternative specifications to ensure robust forecasting performance.

The values of the MAE for each model are:

Table 1: MAE, AIC and BIC for ARIMA Models

Model	MAE	AIC	BIC
$\text{ARIMA}(1, 0, 1)(2, 0, 1)[7]$	19.3057	9555.4786	9596.5992
$\text{SARIMAX}(1, 0, 0)(1, 1, 1)[7]$	17.3733	9544.9475	9574.3193
$\text{SARIMAX}(0,1,2)(1,0,1)[7]$ with dummies	17.6428	9573.3827	9626.2656

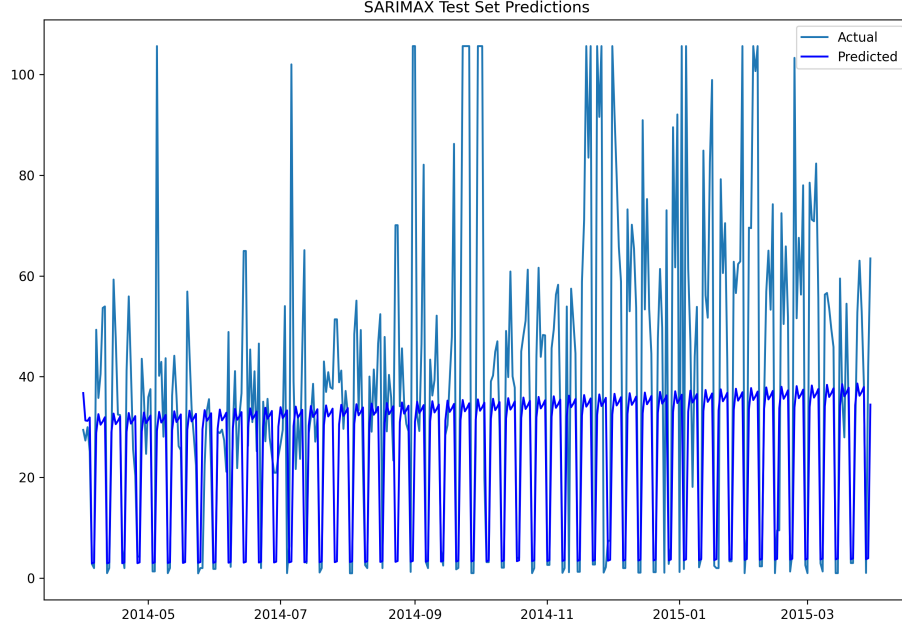


Figure 7: Model 2 predictions on the test set

Table 1 presents the Mean Absolute Error (MAE), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC) for three ARIMA-based models. Model 2 (SARIMAX(1, 0, 0)(1, 1, 1)[7]) demonstrated the best performance with the lowest MAE (17.3733), AIC (9544.95), and BIC (9574.32). This suggests that incorporating seasonal differencing ($D=1$) effectively captured residual non-stationarity while maintaining simplicity with a single non-seasonal autoregressive term ($p=1$) and a seasonal component. By contrast, Model 3 (SARIMAX(0, 1, 2)(1, 0, 1)[7] with dummies), which included weekend and holiday dummies as exogenous variables, achieved a slightly higher MAE (17.6428) and greater model complexity, as indicated by higher AIC (9573.38) and BIC (9626.27). This reflects that while exogenous variables can account for external influences, they did not improve predictive accuracy significantly in this case, possibly due to weak correlations between holidays or weekends and the target variable.

5 UCM

Unobserved Components Models (UCM) are a flexible class of state-space models used for time series analysis and forecasting. Unlike traditional regression-based approaches, UCM explicitly decomposes a time series into interpretable components, including trend, seasonal patterns, cyclical effects, and irregular

noise. This decomposition allows for a detailed understanding of the underlying data structure and facilitates forecasting by modeling each component separately.

The key components of a UCM include:

- **Trend:** Represents the long-term movement in the data. The trend can be specified as a fixed level, a stochastic level, or a local linear trend to capture time-varying dynamics.
- **Seasonality:** Accounts for repeating patterns over a fixed period. UCM allows modeling seasonality using both dummy variables or trigonometric functions (harmonic representation) for greater flexibility.
- **Cycle:** Describes cyclical movements not associated with fixed seasonality. It is often used to model business cycles or economic fluctuations.
- **Exogenous Variables:** Incorporates external predictors or regressors to capture additional effects that influence the time series.

Model parameters are estimated using maximum likelihood methods, and model selection is typically guided by criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), as we've done in this project along with the MAE. UCMs are particularly valuable for handling non-stationary data, irregular seasonalities, and complex trend structures, making them well-suited for time series forecasting tasks.

The first model includes a local linear trend component and a weekly seasonal component modelled using dummy variables used in the previous ARIMA model. This structure captures a time-varying trend and short-term seasonality, accounting for variations in weekly patterns. The inclusion of exogenous variables, such as holiday and weekend dummies, allows the model to incorporate external influences on the series. This model provides a baseline framework for capturing both trend and seasonality in the data.

The second model extends the base framework by adding a frequency-based annual seasonal component, using a period of 365.25 days with two harmonics to capture yearly variations. This approach allows for more refined seasonal adjustments, addressing both weekly cycles and broader annual patterns, which are common in time series influenced by yearly events. This model increases complexity to improve long-term accuracy.

The third model introduces multiple seasonal components, combining a monthly pattern (approximated with a 30.44-day period and three harmonics) and an enhanced weekly component with two harmonics. This configuration is designed to capture interactions between short-term and medium-term cycles.

The selection of these models follows a progressive strategy, beginning with a simple baseline and adding complexity through additional seasonal components and harmonics.

Table 2: MAE, AIC, and BIC for UCM Models

Model	MAE	AIC	BIC
Dummy variables as regressors, weekly seasonal components, and local linear trend	<i>19.0792</i>	<i>9590.5955</i>	<i>9631.7346</i>
Dummy variables as regressors, frequency-based annual seasonal components with two harmonics, and local linear trend	<i>18.7484</i>	<i>9614.0051</i>	<i>9661.0091</i>
Dummy variables as regressors, multiple seasonal components, and local linear trend	18.2852	9589.9278	9642.7867

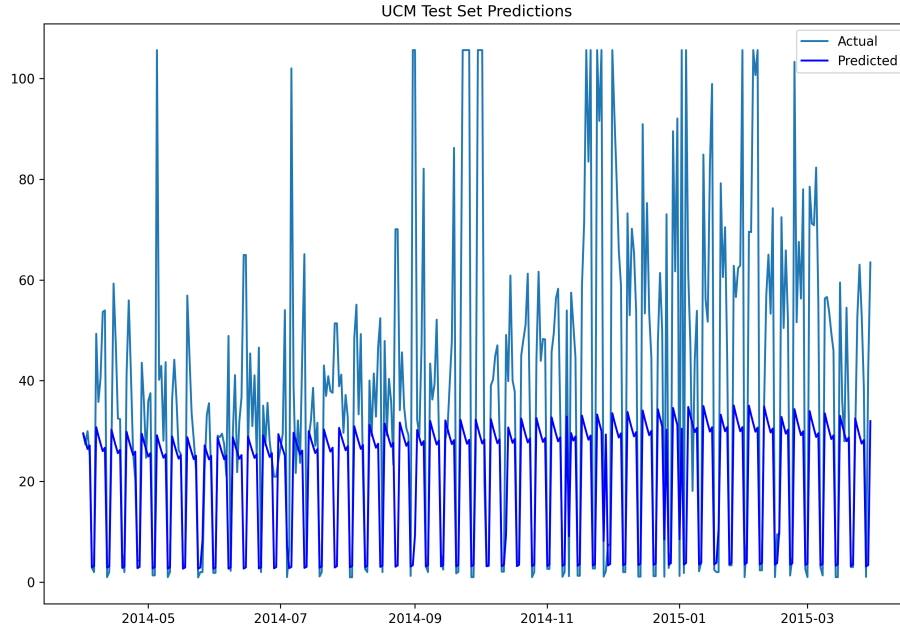


Figure 8: Model 3 predictions on the test set

Table 2 presents the MAE, AIC, and BIC values for three Unobserved Components Models (UCM). Model 3, which includes dummy variables as regressors, multiple seasonal components, and a local linear trend, outperforms the other models across all metrics, with the lowest MAE (18.2852), AIC (9589.93), and BIC (9642.79). This indicates that Model 3 provides the most accurate forecasts while balancing complexity and fit.

Model 1, a simpler model using only weekly seasonal components with dummy variables and a local linear trend, shows slightly higher error (MAE of 19.0792) but achieves a lower AIC and BIC compared to Model 2. Model 2 introduces a frequency-based annual seasonal component with two harmonics, which marginally improves MAE (18.75) but increases AIC (9614.01) and BIC (9661.01), reflecting higher complexity without sufficient accuracy gain.

The superior performance of Model 3 can be attributed to its ability to capture more nuanced seasonal patterns, combining both monthly and weekly cycles. The use of harmonics for each period provides flexibility in modeling short-term and medium-term seasonality, improving overall forecast accuracy while maintaining a relatively balanced model fit, as indicated by its AIC and BIC scores.

6 Machine Learning

Machine Learning (ML) is a subset of artificial intelligence that focuses on developing algorithms capable of learning patterns from data and making predictions or decisions without explicit programming. Unlike traditional statistical models, ML models can capture complex, non-linear relationships and adapt to large datasets with high-dimensional features. The learning process in ML involves training a model on historical data and testing it on unseen data to assess its generalization ability.

ML models are broadly categorized into supervised, unsupervised, and reinforcement learning. In supervised learning, models are trained on labelled data, where the input features are associated with known target values. Time series forecasting, like in this project, falls under supervised learning, where the objective is to predict future values based on past observations and external predictors.

The Random Forest (RF) model is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and control overfitting. Each tree is built on a random subset of the data and features, and the final prediction is made by averaging the outputs of all trees. This approach reduces variance and increases robustness. In this project, the RF model was chosen for its ability to handle non-linear relationships and interactions between predictors, such as quarter, month, and day of the week, which are important for capturing seasonality and calendar effects in the data.

Support Vector Regression (SVR) is a type of Support Vector Machine (SVM) used for regression tasks. SVR finds a function that approximates the data within a margin of tolerance, maximizing generalization. The radial basis function (RBF) kernel was used to capture non-linear relationships, with hyperparameters such as C , γ , and ϵ optimized for flexibility and error control, in our case C is equal to 100, γ and ϵ are equal to 0.1.

XGBoost (Extreme Gradient Boosting) is a powerful ensemble technique that builds trees sequentially, correcting errors from previous trees. It employs regularization to avoid overfitting and utilizes efficient computation for large datasets. Key hyperparameters include learning rate, maximum tree depth, and the number of trees. In this project, XGBoost was chosen for its proven performance in time series tasks, where it efficiently captures both linear and non-linear patterns with tunable complexity.

Table 3: MAE for ML Models

Model	MAE
Random Forest	<i>18.5137</i>
Support Vector Regressor	<i>16.8963</i>
XGBoost	<i>18.3564</i>

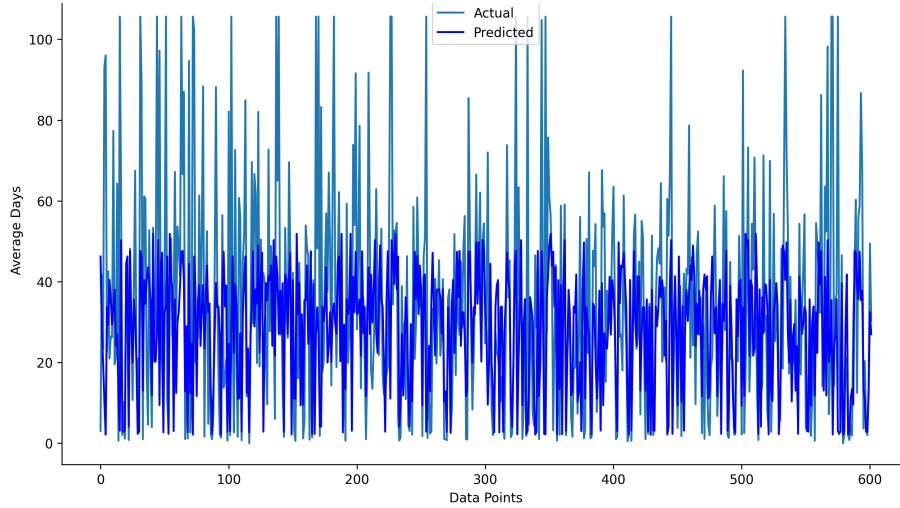


Figure 9: SVR predictions on the test set

Unlike ARIMA and UCM models, Machine Learning models do not rely on likelihood-based estimations, making information criteria such as AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) unusable. Instead, MAE was chosen as the primary evaluation metric because it provides a straightforward measure of the average magnitude of errors between predicted and actual values, without considering the direction of errors.

Table 3 summarizes the Mean Absolute Error (MAE) for the three machine learning models used in this project. Among the models evaluated, the Support Vector Regressor (SVR) achieved the lowest MAE (16.8963), indicating the best predictive accuracy. This superior performance can be attributed to SVR’s ability to handle complex, non-linear relationships between features using the radial basis function (RBF) kernel, which captures subtle patterns that ensemble-based methods like Random Forest and XGBoost may not fully address.

The Random Forest Regressor recorded an MAE of 18.5137. Despite its robustness to overfitting and strong performance in handling categorical and temporal data, its inherent averaging mechanism may have limited its ability to capture fine-grained temporal dependencies in the series. Similarly, XGBoost, while efficient and powerful, achieved a slightly lower MAE of 18.3564. The marginally higher error suggests that the chosen hyperparameters or its handling of interactions in this specific dataset may not have been as effective as the SVR approach.

7 Conclusion and future developments

This project aimed to forecast daily values from April 1, 2015, to November 7, 2015, using three distinct modeling approaches: ARIMA, Unobserved Components Models (UCM), and Machine Learning (ML) models. Extensive pre-processing was conducted to ensure data quality, including handling outliers, transforming the series with a Box-Cox transformation, and incorporating exogenous variables representing weekends and holidays.

Among the ARIMA models, the SARIMAX(1, 0, 0)(1, 1, 1)[7] configuration provided the best performance with a Mean Absolute Error (MAE) of 17.3733, balancing model complexity and predictive accuracy. The inclusion of exogenous variables did not significantly improve performance, highlighting the adequacy of purely seasonal and trend-based components for this dataset.

For UCM, the model incorporating multiple seasonal components with harmonics and a local linear trend achieved the best results, with an MAE of 18.2852, outperforming simpler specifications with only weekly seasonality. This demonstrates the importance of capturing both short-term and medium-term cyclic patterns in complex time series data.

In the machine learning category, the Support Vector Regressor (SVR) model delivered the most accurate forecasts, achieving an MAE of 16.8963. Its kernel-based approach effectively modelled the non-linear dependencies in the data.

For future developments, one promising direction could be integrating real-time updating mechanisms and dynamic regressors that could enhance the adaptability of the models for operational forecasting, ensuring robustness to evolving patterns in the data. Future work may also focus on refining the evaluation process by incorporating additional error metrics and exploring the stability of predictions across different time horizons.